

MOHAMOUD ABUKAR MOHAMOUD

Electronics & Telecommunication Engineer | Software Developer

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PROFESSIONAL SUMMARY

Electronics and Telecommunication Engineering graduate (GPA 3.67) pursuing a Master of Internet Systems at Kigali Independent University (ULK), with thesis research on optimizing federated learning for resource-constrained edge devices in smart grids. Full-stack development experience with TypeScript, React Native, and Node.js, alongside strong foundations in Python, machine learning, IoT systems, and networking. Native Somali speaker, fluent in English, based in Kigali and available immediately. Eager to contribute to software development and digital transformation projects across East Africa.

EDUCATION

Master of Internet Systems (In Progress)

Dec 2024 – Present

Kigali Independent University (ULK) – Kigali, Rwanda | Part-time

- Thesis: "Optimizing Federated Learning for Resource-Constrained Edge Devices in Smart Grids: A Case Study in Somalia", supervised by Dr. Danny Kamuhanda.
- Built a complete federated learning system (Python, Flower, LSTM) deployed on real edge hardware (Raspberry Pi 4, Pi Zero, ESP32), achieving ~50% memory reduction via INT8 quantization, ~60% bandwidth savings via Top-K compression, and ~40% training time reduction with adaptive local epochs.

Bachelor of Science in Electronics and Telecommunication Engineering

2020 – 2025

Amoud University – Borama, Somalia | GPA: 3.67 / 4.0

- Relevant coursework: Programming Fundamentals, Data Communication, Embedded Systems, Digital Signal Processing, Network Design.
- Final-year project: IoT-based plant monitoring system, integrating sensors, microcontrollers, and remote data access.

Secondary Education Certificate

2020

Alif Secondary School – Mogadishu, Somalia

PROFESSIONAL EXPERIENCE

ICT Intern — Somtel Telecommunications

Jun 2022 – Aug 2022

Hargeisa, Somaliland

- Assisted in network troubleshooting and diagnostics across enterprise network infrastructure, contributing to faster issue resolution.
- Configured and maintained network devices, gaining hands-on experience with real-world telecom systems.
- Provided first-line technical support to staff and end users, resolving software installation and connectivity issues.

TECHNICAL PROJECTS

Federated Learning for Smart Grids (Master's Thesis Research)

2025 – Present

- Designed a privacy-preserving machine learning pipeline for household energy forecasting across distributed, low-resource edge devices, targeting Somali smart grid conditions.
- Implemented LSTM models with INT8 quantization, Top-K gradient compression, and adaptive local epochs using the Flower federated learning framework.

- Open-source code: github.com/maxamud123/federated-learning-smart-grid

Real-Time Traffic Monitoring System (Full-Stack)

2025

- Built a cross-platform mobile and web application (React Native, Expo, TypeScript) with a Node.js backend, providing live traffic visualization, congestion metrics, and route planning.
- Implemented real-time data streaming with Socket.IO and interactive mapping with Leaflet, including push alerts categorized by incident type and severity.

Additional Web Projects

2024 – 2025

- Acacia Coffee (TypeScript web application), phishing detection system (TypeScript), and a personal portfolio website. All available at github.com/maxamud123.

IoT Plant Monitoring System (Final-Year Project)

2025

- Designed and built an IoT solution to monitor plant health parameters (soil moisture, temperature, humidity) in real time.
- Integrated sensors with a microcontroller and enabled remote monitoring via a connected interface.

Smart Home Automation System

2024

- Developed an Arduino-based prototype enabling remote control of home appliances through a mobile interface.
- Implemented safety features including smoke detection and motion sensing for automated alerts.

Student Management System

2023

- Built a database-driven application managing student records with full CRUD functionality using Python and MySQL.
- Applied software development lifecycle principles from requirements gathering to testing.

TECHNICAL SKILLS

- Programming: Python, Java, C/C++, TypeScript, JavaScript, HTML, CSS.
- Web & Mobile: React Native, Expo, Node.js, Socket.IO, Leaflet, responsive design.
- Machine Learning: Federated learning (Flower), LSTM models, model quantization and compression for edge devices.
- Tools & Platforms: Git, Visual Studio, Android Studio, Arduino, MATLAB.
- Databases: MySQL, SQLite.
- Networking: TCP/IP, network configuration, wireless communication.
- Other: Circuit design, signal analysis, debugging, software testing.

LANGUAGES

- Somali — Native speaker.
- English — Fluent (written and spoken).
- Arabic — Basic proficiency.